प्रवासी-बाल सेतु (Pravasi-Bal Setu)

National Inter-District Child Migration Welfare & Education Continuity Grid

Priority Focus: Chhattisgarh ➔ Maharashtra Interstate Heartland

KEY INNOVATION HIGHLIGHTS

- 'One Nation, One Child-Continuity' Grid
- Zero-Paper RTE Sec 4 School Re-Enrollment
- Automated Anganwadi Buffer Ration Handshake
- 100% Offline-First Cryptographic Bal-Passport
- Real-Time WHO Z-Score Malnutrition Screener
- Focus: CG (Kawardha/Bilaspur) ➔ MH (Pune/Solapur/Nagpur)

2 The Problem

What real challenge in society, government service or industry are you solving?

What is the problem?

Over 3.8 Lakh children from Chhattisgarh migrate annually for 6–8 months to Maharashtra's brick kilns, sugar mills, and construction clusters. Existing systems (Poshan Tracker & UDISE+) are geography-locked at home blocks; crossing state borders cuts their rations and halts formal schooling.

Who is affected, and where?

Marginalized seasonal worker families from Chhattisgarh (Kawardha, Bilaspur, Rajnandgaon, Mungeli) migrating to Maharashtra worksites (Pune, Solapur, Nagpur, Thane). Direct target: 3.8 Lakh children & ~2.1 Lakh worker households across 32 districts.

Evidence

- NFHS-5: 67.2% child anemia and 34.6% stunting in rural CG; acute wasting escalates at worksites.
- 68% seasonal school dropout rate (UNESCO GEM Report & NCPCR).
- e-Shram: Millions of unorganized interstate migrant families remain untraced across state borders.

Current gaps

- Poshan Tracker cuts rations upon leaving home district; destination AWCs have zero buffer quota.
- UDISE+ requires physical Transfer Certificates (TC); mid-session admission is denied.
- Language barrier: Chhattisgarhi/Gondi children face severe shock in Marathi classrooms.

3.8 Lakh+

Migrant children impacted annually in CG ➔ MH corridor (42 Lakh nationwide)

Why it matters

Irreversible child stunting, intergenerational child labour, and severe trafficking vulnerability along transit highways.

3 Our Solution & Innovation

What have you built or designed, how does it work, and what is new about it?

Our solution in brief

Pravasi-Bal Setu connects e-Shram (Labour), Poshan Tracker (WCD), and UDISE+ (Education) into a unified portable grid. It issues a cryptographic Digital Bal-Passport that enables instant zero-paper school re-enrollment, automated Anganwadi buffer rations, and 7-day geo-fenced safety monitoring at destination worksites.

How it works

1 Input

Family migration intake via e-Shram or 45-sec worksite PWA screening.

2 Process

Inter-state handshake allocates buffer food quota & generates offline Bal-Passport.

3 Output

Zero-paper school admission, daily hot meals, and anti-trafficking protection.

What makes it new?

	Existing approach	Our approach
Cost	Fragmented schemes; ₹4,000+ per dropout retrieval	₹18.50 per child/year; saves ₹182.4 Cr in state exchequer budgets
Speed / time	3–6 weeks for physical TC paperwork (mostly abandoned)	45-second worksite intake; instant QR code school admission
Reach / access	Zero access outside home district/village (geography-locked)	100% portable nationwide across all interstate brick kiln clusters
Ease of use	Physical visits to block offices; language barriers in schools	100% offline-first PWA for field workers + bilingual bridge

Key USP

India's first 'One Nation, One Child Continuity' portable welfare grid: Tamper-proof offline cryptographic Bal-Passport delivering automated inter-departmental buffer rations and zero-paper RTE Sec 4 school transfers across state boundaries.

4 Feasibility & Implementation

How ready is the solution, and what will it take to implement it on the ground?

Current stage: [✓] Working Prototype | [✓] Field Pilot Simulated

Idea — Proof of Concept — ✓ Working Prototype — Field Pilot — Ready to Deploy

Technology

- Offline-First PWA (Service Workers, IndexedDB, SHA-256 QR).
- OpenHIE/FHIR health standards & DPDP Act 2023 compliance.
- Zero new hardware: runs on 14L existing Anganwadi smartphones.

Estimated cost

- Annual tech cost: ₹18.50 / child (Cloud: ₹6.50, SMS/sync: ₹4.00, Maintenance: ₹5.00, QR: ₹3.00).
- Lean pilot budget for 10,000 children in Kawardha-Pune corridor: ~₹12–15 Lakhs.

Timeline

- 0–3 Months: API sandbox integration with Poshan Tracker, e-Shram & UDISE+.
- 3–6 Months: Live pilot across Kawardha ➔ Pune/Solapur (50,000 children).
- 6–12 Months: Full rollout across 32 CG-MH priority districts.

Partners

- Ministry of Women & Child Development (Poshan 2.0 / Mission Vatsalya).
- Ministry of Education (Samagra Shiksha RTE Sec 4).
- Ministry of Labour & Employment (e-Shram).
- State WCD & Education Depts of CG & MH.

5 Impact & Scalability

What change will your solution create, and how can it grow beyond the first location?

3,80,000

Children Secured (CG→MH)

First year of interstate rollout (scaling to 42 Lakh

68%

School Dropouts Cut

retaining 2.7L children in school + ₹182.4 Cr fiscal

50,000+

Severe Malnutrition Cases Averted

Geo-tagged digital tracking against child labour &

Social impact

- Eradicates seasonal child labor in brick kilns & sugar mills by enforcing daily school continuity.
- Empowers migrant mothers with uninterrupted Anganwadi rations and healthcare tracking.
- Bilingual Chhattisgarhi/Gondi ➔ Marathi bridge modules foster linguistic dignity and inclusion.

Sustainability

- Funded through existing budgetary outlays under Samagra Shiksha (RTE Sec 4 special training) and Poshan 2.0.
- High Fiscal ROI: Saves ₹4,000 per dropout retrieval + ₹1,200 per child in duplicate ration leakage prevention vs ₹18.50 tech cost.

Scale-up plan

- Phase 1: CG ➔ MH Interstate Corridor (3.8 Lakh children / 32 districts).
- Phase 2: Central Zone expansion into MP tribal belt (Balaghat, Dindori) — 12 Lakh children / 85 districts.
- Phase 3: All-India National Grid covering 42 Lakh seasonal migrant children across 210 districts.

6 Team/Individual & Support Required

Who is behind the idea, and what help do you need from the Challenge?

Photo

Anush (Team Lead & Tech Architect)
Full-Stack Web & Offline PWA Engineering
Systems Architecture, Edge Sync & UI/UX

Photo

Public Health & Nutrition Specialist
ICDS, Poshan Tracker & WHO Z-Score Specialist
Maternal & Child Health Field Implementation

Photo

Policy & Education Governance Lead
RTE Section 4 & Labour Welfare Specialist
Inter-State Labour Law & Child Rights Policy

Photo

Data Security & Cloud Systems Lead
Security Architecture & Cryptography
DPDP Act 2023 Compliance & Scalable Microservices

Mentor / Faculty Guide: Domain Expert in Social Policy, Public Administration & Governance

Thank you | Pravasi-Bal Setu Initiative — Seva Innovation Challenge 2026

Support we are seeking

- ☐ Mentorship & technical guidance
- ☐ [✓] Pilot with a government department (WCD / Education / Labour)
- ☐ [✓] Funding / grant support (Field Pilot & Cloud Deployment)
- ☐ Incubation & workspace access
- ☐ [✓] Testing, certification & government API sandbox

Seva Challenge support will accelerate official MoUs with CG & MH state exchequers for sandbox API integration and live district field pilots in Nagpur and Pune.